

1 Solution — GIAM 7.3

Problem 4

1.1 Problem

Given any set of **53** integers, show that there are two of them having the property that either their sum or their difference is evenly divisible by **103**.

1.2 Pigeonhole Framing

Working modulo **103**, recall:

$$103 \mid (a - b) \iff a \equiv b \pmod{103}, \quad 103 \mid (a + b) \iff a \equiv -b \pmod{103}.$$

So the question is whether, among **53** integers, *some* pair has residues that are either *equal* or *negatives* of each other modulo **103**. This suggests grouping residues by the equivalence relation

$$r \sim s \iff r \equiv \pm s \pmod{103}.$$

Pigeonholes. The equivalence classes of $\sim$ on $\mathbb{Z}/103\mathbb{Z}$. **Pigeons.** The **53** given integers, each assigned to the class of its residue mod **103**.

If we can count the classes and find their number to be less than **53**, the pigeonhole principle hands us two integers in the same class — and unpacking $\sim$ will give the desired sum-or-difference divisibility.

1.3 Step 1 — Count the Equivalence Classes

The set of residues is $\mathbb{Z}/103\mathbb{Z} = \{0, 1, 2, \dots, 102\}$. Under $\sim$:

- **Singleton class:** $\{0\}$. Reason: $-0 \equiv 0 \pmod{103}$, so **0** pairs with itself.
- **Pair classes:** $\{r, 103 - r\}$ for $r \in \{1, 2, \dots, 51\}$. Each is a *genuine* pair (size **2**) since $r = 103 - r$ would force $2r = 103$ — impossible, as **103** is odd.

For $r = 52, 53, \dots, 102$, the class $\{r, 103 - r\}$ has already been listed (with the roles of r and $103 - r$ swapped). So the *distinct* classes are exactly

$$\{0\}, \quad \{1, 102\}, \quad \{2, 101\}, \quad \{3, 100\}, \quad \dots, \quad \{51, 52\}.$$

Counting:

$$\#\text{classes} = 1 + 51 = 52,$$

and covering $1 + 51 \cdot 2 = 103$ residues, as required. ✓

Why oddness matters. *The proof's count came out to 52 because 103 is odd: there is no residue r with $103 - r = r$, so every nonzero class is a genuine pair. For even N , the residue $N/2$ is its own negative and forms an extra singleton — see the table at the end.*

1.4 Step 2 — A Folded-Residue Map

Concretely, define the *folded residue*

$$g : \mathbb{Z} \longrightarrow \{0, 1, 2, \dots, 51\}, \quad g(n) = \min(n \bmod 103, 103 - (n \bmod 103)),$$

which sends each integer to the *smaller* of its residue and its negative residue. By construction,

$$g(a) = g(b) \iff a \equiv \pm b \pmod{103},$$

so the level sets of g are precisely the 52 equivalence classes from Step 1.

1.5 Step 3 — Apply the Pigeonhole

Let $S = \{n_1, n_2, \dots, n_{53}\}$ be the 53 given integers, and restrict g to S :

$$g|_S : S \longrightarrow \{0, 1, 2, \dots, 51\}.$$

Since

$$|S| = 53 > 52 = |\text{Rng}(g)|,$$

Theorem 7.3.1 tells us $g|_S$ is not injective: there exist distinct $n_i, n_j \in S$ with

$$g(n_i) = g(n_j),$$

i.e.,

$$n_i \equiv \pm n_j \pmod{103}.$$

1.6 Step 4 — Translate to “Sum or Difference Divisible by 103”

Two cases, depending on the sign:

Case A — same sign. $n_i \equiv n_j \pmod{103}$. Then

$$n_i - n_j \equiv 0 \pmod{103} \implies 103 \mid (n_i - n_j).$$

Case B — opposite sign. $n_i \equiv -n_j \pmod{103}$. Then

$$n_i + n_j \equiv 0 \pmod{103} \implies 103 \mid (n_i + n_j).$$

In either case, 103 divides *either* the sum or the difference of two of the given integers.

■

Degenerate sub-case. If $g(n_i) = g(n_j) = 0$, then both $n_i, n_j \equiv 0 \pmod{103}$, and the two cases overlap — 103 divides both the sum and the difference.

1.7 Sharpness — 52 Integers Need Not Suffice

The threshold 53 is *tight*: there is a 52-element set of integers in which **no** two have a sum or difference divisible by 103.

Take

$$S_{\text{bad}} = \{0, 1, 2, \dots, 51\}, \quad |S_{\text{bad}}| = 52$$

— one representative from each of the 52 equivalence classes, namely the *smaller* element of each class. For distinct $a, b \in S_{\text{bad}}$ with $a < b$:

- **Difference:** $1 \leq b - a \leq 51 < 103$, so $103 \nmid (b - a)$.
- **Sum:** $1 \leq a + b \leq 50 + 51 = 101 < 103$, so $103 \nmid (a + b)$.

So $\mathcal{S}_{\text{bad}}$ avoids the desired property entirely — confirming that the pigeonhole threshold of **53** cannot be lowered.

1.8 Remark — The General Form

The argument used only one fact about **103**: that it is odd. For an arbitrary integer $N \geq 2$, the *fold-by-negation* equivalence

$$r \sim s \iff r \equiv \pm s \pmod{N}$$

partitions $\mathbb{Z}/N\mathbb{Z}$ into:

- The singleton $\{0\}$.
- (If N is even) the singleton $\{N/2\}$, since $N/2 \equiv -N/2 \pmod{N}$.
- Pair classes $\{r, N - r\}$ for r in the appropriate range.

This gives

$$\# \text{classes} = \begin{cases} \frac{N+1}{2}, & N \text{ odd,} \\ \frac{N+2}{2}, & N \text{ even.} \end{cases}$$

Pigeonhole then yields:

Any collection of $\# \text{classes} + 1$ integers contains two whose sum or difference is divisible by N .

N	Parity	$\#$ classes	Pigeonhole threshold
3	odd	2	3
4	even	3	4
5	odd	3	4
10	even	6	7
103	odd	52	53
general odd N	—	$(N+1)/2$	$(N+3)/2$

general even N	—	$(N + 2)/2$	$(N + 4)/2$
------------------	---	-------------	-------------

Table 1.1

Each threshold is tight, witnessed by taking one representative from each class (e.g., $\{0, 1, 2, \dots, \lfloor N/2 \rfloor\}$).

1.9 Remark — A Small Verification

Take $N = 5$, threshold 4. Classes: $\{0\}$, $\{1, 4\}$, $\{2, 3\}$ — three of them.

- **Bad set of size 3:** $\{0, 1, 2\}$. Sums: 1, 2, 3; differences: 1, 2. None divisible by 5. ✓
- **Any set of size 4:** by pigeonhole, two must share a class. E.g., $\{0, 1, 2, 3\}$: residues 2, 3 are in class $\{2, 3\}$, and $2 + 3 = 5$. ✓ (Or $\{0, 1, 2, 4\}$: residues 1, 4 in class $\{1, 4\}$, and $1 + 4 = 5$. ✓)

The $N = 103$ instance follows the same template — just at a larger scale.

1.10 Remark — The Whole Idea is the Right Equivalence

As with the previous problem in this section, the heart of the proof is *choosing the equivalence relation*. Once we group residues by $r \sim -r$, the pigeonhole count comes out to 52, the threshold 53 is one above that, and translating “same class” back into “sum or difference $\equiv 0$ ” is immediate.

A direct attack — checking all $\binom{53}{2} = 1378$ pairs — gives no useful structure. The folded residue g is the *minimal* invariant such that “ $g(a) = g(b)$ ” is equivalent to “ $103 \mid (a \pm b)$.” Naming that invariant *is* the proof.